Lab 5: Average Treatment Effect

Causal Machine Learning

Stijn Vansteelandt and Federico Bertoia, Ghent University

We will be using data from an observational study of 996 patients receiving a Percutaneous Coronary Intervention (PCI) with/without augmented therapy with abcximab (a IIb/IIIa cascade blocker) at Ohio Heart Health in 1997 and the patients are followed for at least 6 months by the staff of the Lindner Center. The outcome of interest is whether they survived at least 6 months. Note that patients receiving abciximab tended to be more severely diseased than those who did not.

1. Carry out a naive (unadjusted) analysis to estimate the marginal association between treatment and outcome in terms of a risk difference (RD) and/or odds ratio (OR).

2. Use G-computation to estimate the average treatment effect (ATE). Try in the following ways:

   (a) Estimate the ATE using a parametric model for the nuisance parameters. Interpret your results. Note how much simpler interpretation is relative to question 1.

   (b) You can use the R package `stdReg` to obtain the standard error.

   (c) What are the advantages and disadvantages of using parametric models for G-computation?

   (d) Estimate the ATE, now using the R package `SuperLearner` to estimate nonparametrically the nuisance parameters.

   (e) What are the advantages and disadvantages of using nonparametric machine learning algorithms for G-computation?

   (f) Do you see ways to obtain an accompanying standard error?

3. Use Inverse Probability Weighting (IPW) to estimate the average treatment effect. Try in the following ways:

   (a) Estimate the ATE using a parametric model for the propensity scores. Note that you can also use a weighted linear regression for this.

   (b) Estimate the ATE, now using the R package `SuperLearner` to estimate nonparametrically the propensity scores and then both a weighted linear regression and the Horvitz-Thompson estimator.

   (c) Check the distribution of the estimated propensity scores as well as the inverse probability weights for the 2 treatment groups. Is there evidence of positivity violations?

4. Use Augmented Inverse Probability Weighting (AIPW) to estimate the average treatment effect. Try in the following ways:

   (a) Estimate the ATE using the outcome model and propensity score estimates obtained in the previous steps.

   (b) Compute the standard error based on the ATE's efficient influence curve.

(c) Now estimate the ATE using the R package `npcausal` or the function `est_aipw`. Change the number of folds (e.g., 1, 2, 5) and evaluate how that impacts results.

(d) Which are the advantages of AIPW with relative G-computation and IPW? Do you see any drawback?

5. Use Targeted Maximum Likelihood to estimate the average treatment effect. Try in the following ways:

   (a) Estimate the ATE using `SuperLearner` to estimate the nuisance parameters and using the fluctuation model of the form

   $$\operatorname{logit} E(Y \mid A, L) = \operatorname{logit} \bar{Q}_n^{(0)}(L) + \delta \frac{A}{g_n(L)}$$

   (b) Compute the standard error and 95% confidence intervals?

   (c) Estimate the ATE, now using the R package `tmle`. Evaluate if you find different results with a more extensive library of learners to use in the `SL.library` argument of the `SuperLearner` package.

(bonus) If you have time left, you can repeat question 4 and 5, now focussing on the average treatment effect in the treated (ATT).